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4 SEM BCA (CBCS) DCCN 4-3

2022

(July)

COMPUTER APPLICATION

Paper : 4-3

**(Data Communications and
Computer Networks)**

Full Marks : 60

Time : Three hours

***The figures in the margin indicate
full marks for the questions.***

1. Answer the following : 5
- (a) The application layer of the OSI model is
- (i) four layer
 - (ii) five layer
 - (iii) six layer
 - (iv) seven layer

Contd.

(b) A device which can be connected to a network without using cable is called

- (i) distributed device
- (ii) centralized device
- (iii) open-source device
- (iv) wireless device

(c) Physical layer is concerned with transmitting raw bits over a communication channel. (True/False)

(d) In data communication components, sender is a transmitting device. (True/False)

(e) The network layer is concerned with the controlling of operation of the subnet. (True/False)

2. (a) Write the functionalities of transport layer and network layer of ISO/OSI model. 5

OR

(b) Write the functionalities of application layer and presentation layer of ISO/OSI model.

3. Compare and contrast **any one** of the following : 5

(a) Parallel and serial communications

(b) Synchronous and asynchronous communications

4. (a) Briefly explain simplex, half duplex and full duplex communications $2 \times 3 = 6$

(b) How does TDM differ from FDM? 4

5. (a) What is the significance of twist in the twisted pair cable? What are different types of twisted pair cables? $2 + 2 = 4$

(b) What are the advantages and disadvantages of fiber optic cables? 6

OR

(c) Compare between twisted pair cable and coaxial cable with suitable diagrams.

6 (a) Briefly explain BUS topology and STAR topology with suitable diagrams. 5

OR

- (b) Briefly explain MESH topology and RING topology.
7. (a) Briefly explain the working principle of **any one** of the following : 6
- (i) Packet switching
 - (ii) Circuit switching
- (b) What is a virtual circuit ? 4
8. (a) Briefly explain **any one** of the following : 8
- (i) Distance vector routing protocol
 - (ii) Link state routing protocol
- (b) What is an IP address ? What are different classes of IP address ? 2
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